

Form Claw — Software Requirements Specification (SRS)

Version: 2.0
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Project: Form Claw — Automated Hebrew PDF Form Filler

1. Introduction

1.1 Purpose

Form Claw automates the process of filling Hebrew PDF forms for the Russak family. Users send a form via email and receive the completed form as a reply, eliminating manual data entry.

1.2 Scope

- The system encompasses:
- Email reception and attachment processing
 - AI-powered form analysis (field detection, coordinate mapping)
 - Automated form filling (text, signatures, checkboxes, ID digits)
 - Reply delivery with the filled PDF
 - Monitoring dashboard with analytics and health checks

1.3 Definitions

Term	Definition
Form Claw	The complete automated form-filling system
Processor	The Cloud Run service that performs AI analysis and form filling
Email Worker	The Cloudflare Worker that receives and routes emails
Knowledge Base	Family-specific data entries stored in Firestore
Intake Drop	An email received but not processed (no valid attachments)

2. Overall Description

2.1 System Context

Form Claw operates as an email-driven automation pipeline:

- 1. Family members send emails with PDF or image attachments to `formclaw@savlil.com`
- 2. The system identifies the target person and signer from email context
- 3. AI vision analyzes form structure and generates filling code
- 4. The filled PDF is sent back as an email reply

2.2 User Classes

User	Description	Access
Family member	Sends forms for filling	Email (whitelisted)
Admin	Monitors system, manages knowledge	Dashboard (Google SSO)

2.3 Operating Environment

Component	Environment
Email reception	Cloudflare Email Routing
Email processing	Cloudflare Workers (V8 isolate)
Form processing	Google Cloud Run (container)
Dashboard	Google Cloud Run (container)
Data storage	Google Cloud Firestore
File storage	Google Cloud Storage
Email sending	Resend API
AI model	Google Gemini Flash

3. Functional Requirements

3.1 Email Reception (FR-100)

ID	Requirement	Priority
FR-101	System SHALL receive emails at formclaw@savlil.com	Must
FR-102	System SHALL accept PDF attachments (application/pdf)	Must
FR-103	System SHALL accept image attachments (JPG, PNG, WEBP, HEIC)	Must
FR-104	System SHALL convert image attachments to PDF before processing	Must
FR-105	System SHALL reply with an error when no valid attachments are found	Must
FR-106	System SHALL extract email threading headers for proper reply threading	Should
FR-107	System SHALL classify attachments by MIME type and file extension	Must

3.2 Form Analysis (FR-200)

ID	Requirement	Priority
FR-201	System SHALL convert PDF pages to images for AI analysis	Must
FR-202	System SHALL identify all fillable fields with coordinates	Must
FR-203	System SHALL detect field types: text, date, checkbox, radio, signature, ID digits	Must
FR-204	System SHALL identify OR/slash selection constructs (כן/לא, אב/אם)	Must
FR-205	System SHALL determine the target person from email context	Must
FR-206	System SHALL determine the signer (parent) from email context	Should
FR-207	System SHALL identify the form's purpose and issuing body	Should

3.3 Form Filling (FR-300)

ID	Requirement	Priority
FR-301	System SHALL generate Python code using ReportLab for PDF overlay	Must
FR-302	System SHALL handle Hebrew RTL text with character reversal	Must
FR-303	System SHALL fill ID number digit boxes individually	Must
FR-304	System SHALL draw ellipses around selected options (OR/slash)	Must
FR-305	System SHALL overlay signature PNG images with transparency	Must
FR-306	System SHALL prevent text overflow beyond field boundaries	Must
FR-307	System SHALL use family data and knowledge base entries	Must
FR-308	System SHALL merge overlay with original PDF without quality loss	Must
FR-309	System SHALL support multi-page forms	Must

3.4 Reply Delivery (FR-400)

ID	Requirement	Priority
FR-401	System SHALL reply to the sender with the filled PDF attached	Must
FR-402	System SHALL thread the reply to the original email	Should
FR-403	System SHALL send error notifications on processing failure	Must
FR-404	System SHALL send explanatory replies for intake drops	Must

3.5 Security (FR-500)

ID	Requirement	Priority
FR-501	System SHALL authenticate webhook requests with bearer token	Must
FR-502	System SHALL scan email content for prompt injection attacks	Must
FR-503	System SHALL detect Hebrew and English injection patterns	Must
FR-504	System SHALL block emails exceeding risk score threshold	Must
FR-505	System SHALL execute generated code in a restricted namespace	Must
FR-506	System SHALL rewrite file paths in generated code to safe locations	Must

3.6 Dashboard (FR-600)

ID	Requirement	Priority
FR-601	Dashboard SHALL require Google SSO authentication	Must
FR-602	Dashboard SHALL display processing activity log	Must
FR-603	Dashboard SHALL show processing statistics and charts	Should
FR-604	Dashboard SHALL provide error tracking with CSV export	Should
FR-605	Dashboard SHALL allow knowledge base management (CRUD)	Must
FR-606	Dashboard SHALL show system health status	Must
FR-607	Dashboard SHALL display Cloudflare email intake analytics	Should
FR-608	Dashboard SHALL provide E2E pipeline health tests	Should

3.7 Logging & Monitoring (FR-700)

ID	Requirement	Priority
FR-701	System SHALL log every processing attempt to Firestore	Must
FR-702	Logs SHALL include: sender, subject, status, timing, target person	Must
FR-703	System SHALL store filled PDFs in Cloud Storage	Must
FR-704	System SHALL track processing time for each request	Should

4. Non-Functional Requirements

4.1 Performance

ID	Requirement	Target
NFR-101	Form processing time	< 60 seconds
NFR-102	Email worker response time	< 5 seconds
NFR-103	Dashboard page load time	< 3 seconds

4.2 Reliability

ID	Requirement	Target
NFR-201	System availability	99.5%
NFR-202	Graceful error handling	Always reply to sender
NFR-203	Post-deploy E2E smoke tests	On every deployment

4.3 Scalability

ID	Requirement	Target
NFR-301	Concurrent processing	Up to 3 instances
NFR-302	Cold start time	< 30 seconds
NFR-303	Scale to zero when idle	Yes

4.4 Security

ID	Requirement	Target
NFR-401	Webhook authentication	Bearer token
NFR-402	Dashboard authentication	Google SSO
NFR-403	Prompt injection detection	Block at $\geq 70\%$ risk
NFR-404	Code execution isolation	Restricted namespace

5. Data Requirements

5.1 Family Data (family_data.json)

- Family members: name (Hebrew/English), ID, birth date, contact info
- Address: street, city, ZIP (Hebrew/English)
- Medical: diagnosis, doctor, HMO
- School information per child
- Instruction parsing keywords (Hebrew/English)

5.2 Knowledge Base (Firestore)

- Key-value entries per person or family-wide
- Categories: medical, educational, administrative
- CRUD operations via dashboard

5.3 Processing Logs (Firestore)

- One document per processing attempt
 - Fields: sender, subject, status, target person, timing, error details
 - Filled PDF path in GCS
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6. Interface Requirements

6.1 Email Interface

- **Input:** Email with PDF or image attachment(s)
- **Supported formats:** PDF, JPG, PNG, WEBP, HEIC
- **Instructions:** Subject line and/or body text in Hebrew or English
- **Output:** Reply email with filled PDF attachment

6.2 Dashboard Interface

- **URL:** <https://formclaw.savlil.com>
- **Auth:** Google SSO (whitelisted accounts)
- **Pages:** Dashboard, Activity, Statistics, Errors, Knowledge, System
- **Responsive:** Desktop-optimized

6.3 Processor API

- **Endpoint:** POST /webhook
- **Auth:** Bearer token
- **Health:** GET /health
- **Format:** JSON request/response